

Verification & Validation Plan - Revision 3

1. Verification Targets

- Registration CLI correctness
- SQLite persistence and transaction safety
- OpenClaw cron watchdog execution
- OpenClaw message send notification path
- State transition correctness including unknown threshold
- Duplicate suppression and concurrency safety

2. Positive Scenarios

1. Register process-backed task with owned marker path; mark completion; verify completion notification.
2. Register long-running task; allow reminder interval to elapse; verify overdue reminder.
3. Register task; mark failure; verify failure notification.
4. Restart OpenClaw; verify task remains tracked and next reminder still occurs.

3. Negative / Stress Scenarios

1. Duplicate task ID registration rejected.
2. Marker file disappears unexpectedly; task transitions to unknown after 2 cycles.
3. Concurrent registration and watchdog cycle do not corrupt the database.
4. Emitter command fails; event log records failure without false delivery claim.
5. Multiple tasks become due in the same cycle.

4. Proof Expectations

Claim	Evidence
Registration	CLI output + SQLite row + task event

succeeded

Watchdog ran

OpenClaw cron job record + run log

Notification sent

openc law message send exit status + event row +
chat target evidence

Unknown threshold
honored

Two consecutive failed observations before unknown
transition

Concurrency safe

Contention test output with intact DB state

5. Exit Criteria

- Emitter path proven using the documented OpenClaw command.
- OpenClaw cron path proven as the scheduler.
- Unknown threshold behavior proven at N=2 cycles.
- At least one negative and one concurrency case proven.
- No false completion or false delivered notification in tested scenarios.